CHAPTER 14

Random variables, concentration and the law of large numbers

S62, S64, S73, S74, S77

Objectives

Calculate the expected value and variance of a sum and of a mean; apply the Bienaymé–Chebyshev inequality; determine a sample size.

Course notes

Sums and transformations

For a finite random variable X, taking the values x_i with probabilities p_i,

$$E(X) = \sum_i x_i p_i, \qquad V(X) = \sum_i (x_i - E(X))^2 p_i = E(X^2) - E(X)^2.$$

The standard deviation is $\sigma(X) = \sqrt{V(X)}$. Expanding the square in $V(aX+b)$, the shift b cancels and the factor a is squared. For finite random variables X, Y and real numbers a, b,

$$E(X+Y) = E(X) + E(Y), \quad E(aX+b) = aE(X) + b, \quad V(aX+b) = a^2 V(X).$$

The linearity of the expected value does not require independence. If X, Y are independent, $V(X+Y) = V(X) + V(Y)$ (accepted without proof). Without this assumption, this formula cannot be applied. The standard deviation of $aX+b$ is $|a|\sigma(X)$.

Example of dependence. If $Y = X$, then $V(X+Y) = V(2X) = 4V(X)$, and not $2V(X)$ as soon as $V(X) \neq 0$.

Proof for the binomial distribution. Write $X = B_1 + \cdots + B_n$ with the independent indicators of the successes. Each B_i satisfies $E(B_i) = p$, $E(B_i^2) = p$ and $V(B_i) = p - p^2$. By linearity and additivity under independence, $E(X) = np$, $V(X) = np(1-p)$.

Sample and mean

A sample of size n of a distribution is a family of n independent random variables with the same distribution. If they have expected value μ and variance σ^2, we write $S_n = \sum_{i=1}^n X_i$ and $M_n = \frac{S_n}{n}$.

Sum and mean

$$E(S_n) = n\mu, \quad V(S_n) = n\sigma^2, \qquad E(M_n) = \mu, \quad V(M_n) = \frac{\sigma^2}{n}, \quad \sigma(M_n) = \frac{\sigma}{\sqrt{n}}.$$

To divide the standard deviation of the mean by 2, multiply the size by 4. The variance of the sum increases; that of the mean decreases.

Concentration

Bienaymé–Chebyshev inequality

For a random variable with expected value μ, variance σ^2, and $\delta > 0$,

$$P(|X - \mu| \geq \delta) \leq \frac{\sigma^2}{\delta^2}.$$

An upper bound greater than 1 remains true, but gives no further information; it can be replaced by 1.

Proof in the finite case. In $V(X) = \sum_x (x - \mu)^2 P(X = x)$, keep only the values satisfying $|x - \mu| \geq \delta$. Each square is then at least δ^2. Hence $V(X) \geq \delta^2 P(|X - \mu| \geq \delta)$.

Law of large numbers

Applied to the mean of a sample,

$$P(|M_n - \mu| \geq \delta) \leq \frac{\sigma^2}{n\delta^2} \xrightarrow[n\to\infty]{} 0.$$

This is the law of large numbers in the setting considered: for a fixed precision, the probability of a deviation at least equal to this precision tends to zero. This result does not say that each realisation approaches μ monotonically.

Choosing a sufficient size

To guarantee $P(|M_n - \mu| < \delta) \geq 1 - \alpha$, it suffices to choose

$$n \geq \frac{\sigma^2}{\alpha\delta^2}.$$

Take the ceiling. This gives the smallest integer satisfying *this bound*, not necessarily the minimum size for the exact probability.

For a frequency of successes, $\mu = p$ and $\sigma^2 = p(1 - p) \leq \frac{1}{4}$. If p is unknown, $n \geq \frac{1}{4\alpha\delta^2}$ is a uniform guarantee.

Research activity

Fluctuations of a frequency — 40 min.

A fair coin is tossed independently. For every integer $n \geq 1$, let F_n be the frequency of "heads" among the first n tosses.

1. Write a program that simulates 1000 tosses, then plot the points $(n; F_n)$ for $1 \leq n \leq 1000$. Does the deviation from $\frac{1}{2}$ necessarily decrease at every toss?
2. Express $E(F_n)$ and $V(F_n)$ in terms of n.
3. Using Bienaymé–Chebyshev, find an upper bound for $P(|F_{1000} - \frac{1}{2}| \geq 0.05)$.
4. Determine an integer n sufficient for $P(|F_n - \frac{1}{2}| < 0.05) \geq 0.99$.
5. State what the simulated trajectory shows and what the inequality obtained guarantees. Does a single simulation constitute a proof of this guarantee?

Progressive exercises

E1 · Practice. Let X be a random variable such that $E(X) = 3$ and $V(X) = 2$. Let $Y = 4X - 1$. Calculate the expected value, the variance and the standard deviation of Y.

E2 · Practice. The random variables X and Y are independent and satisfy $E(X) = 2$, $E(Y) = 5$, $V(X) = 1$ and $V(Y) = 4$. Calculate $E(2X - Y)$ and $V(2X - Y)$.

E3 · Practice. Consider 64 independent random variables with the same distribution, with expected value 10 and standard deviation 4. Calculate the expected value and the standard deviation of their mean.

E4 · Practice. Let X be a random variable such that $E(X) = 100$ and $V(X) = 9$. Using the Bienaymé–Chebyshev inequality, give an upper bound for $P(|X - 100| \geq 6)$.

E5 · Practice. Let M_n be the mean of n independent random variables with the same distribution, with expected value μ and variance 4. Determine a size n sufficient for $P(|M_n - \mu| < 0.2) \geq 0.95$.

Textbook or past-paper work

Baccalauréat, Metropolitan France, 17 June 2025: exercise 1, question 6 (p. 3). At home after S77. Read the context of question 5 and use the expected value and variance values accepted in question 6. The paper is available here: official past paper. Use the data and instructions of this past paper.

Homework

Homework 14 — Quality of an estimate — 55 min.

We repeat $n \geq 1$ independent Bernoulli trials with the same parameter $p \in [0;1]$, which is unknown. Let F_n be the frequency of successes.

1. Prove that $E(F_n) = p$ and $V(F_n) = \frac{p(1-p)}{n}$.
2. Show that $p(1-p) \leq \frac{1}{4}$ for all $p \in [0;1]$.
3. Deduce a size n sufficient for $P(|F_n - p| < 0.02) \geq 0.95$, whatever the value of p.
4. We now know that $p \leq 0.1$. Determine a sharper upper bound for $p(1-p)$, then improve the sufficient size obtained in question 3.
5. For this question only, the n results are all equal to the same Bernoulli variable B with parameter p. Calculate F_n and its variance. Explain why the previous guarantee no longer applies.